



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Centre for Professional and
Continuing Education

(SCTP) Advanced Professional Certificate **Data Science and AI**



Module 3 · Machine Learning & GenAI

L01

How Machines *Learn*.

Sarah's Week at NorthStar Retail

I. A BRIDGE FROM M1 → M2 → M3

Same toolkit. Fundamentally new question.

M1 · Data Analytics

Summarise the past.

Describe data and explain what already happened..

SARAH

"Customer reviews dropped 12% last month."

M2 · Data Engineering

Move & store data reliably.

Build stable data pipes between different systems.

SARAH

"New reviews come to our data warehouse every night."

We Are Here Now

M3 · Machine Learning

Predict the unseen.

Label or forecast things no one has seen yet.

SARAH

"Label 10,000 reviews — positive or negative."

Recap on Terminology

Artificial Intelligence

Artificial Intelligence (AI) is an *umbrella term* for any technology where a computer program is attempting tasks that come naturally to the human brain.

Machine Learning

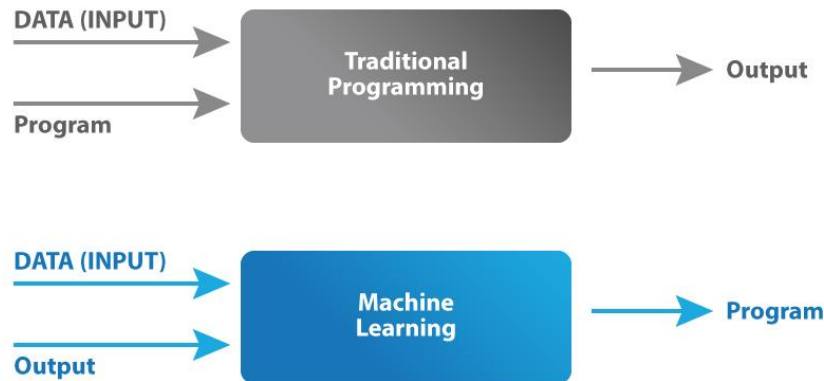
Machine Learning (ML) is a *subset of AI* that involves training algorithms to make predictions or decisions based on data.

Data Science

Data Science is an *interdisciplinary field* that uses statistics & ML methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.

Introduction to Machine Learning

Machine Learning (ML) is a subset of artificial intelligence that provides systems the ability to *automatically learn and improve from experience* without being explicitly programmed.



III. A NORTHSTAR RETAIL CASE STUDY · WEEK 2

Meet Sarah Chen.

Customer Experience Analyst, NorthStar Retail — our protagonist for all of M3.



Sarah Chen

Customer Experience Analyst

NORTHSTAR RETAIL · 2ND WEEK

*Smart, motivated, a little nervous.
First real ML task this week.*

HER FIRST BIG ASK, DROPPED ON HER DESK:

*"By the end of the day, I need to know which of these
10,000 customer reviews are positive and which are
negative.*

Hand-reading them would take a full work-week.

Figure something out."

— Priya · Sarah's manager

V. THE WAY PROGRAMMING WORKED BEFORE ML

Approach 1 — Rule-based programming.

THE PLAN

- 1 List **positive** words: love, perfect, great, nice...
- 2 List **negative** words: terrible, cheap, bad, awful...
- 3 For each review, count hits on each list.
- 4 Whichever count is bigger wins.

TRAITS OF RULE-BASED CODE

- ✓ You write all the rules
- ✓ Computer follows them exactly
- ✓ Deterministic: same input, same output
- ✗ Only handles cases the rules covered
- ✗ Maintenance is whack-a-mole
- ✗ No notion of 'confidence'

VII. THREE FAILURE MODES — AND THEY COMPOUND

Three ways rules quietly break.

#1

No understanding of context

“Terrible shipping — product itself is fine.”

Rules see ‘terrible’ and ‘fine’ but don’t know what each word refers to.

Result: **Incorrectly tagged NEGATIVE.**

#2

Negation flips the original meaning

“Not what I expected — actually really nice in person!”

‘not’ inverts the sentiment of the next phrase. Word-counting misses this.

Result: **Rules tag NEUTRAL or NEGATIVE.**

#3

Endless maintenance work

Every new review brings new edge cases you didn’t plan for.

Your rule list grows 10 → 100 → 1,000 — and conflicts start.

Result: **Maintenance cost balloons.**

VIII. THE PARADIGM FLIP

Stop writing rules. Start providing examples.

TRADITIONAL PROGRAMMING

**You write the rules.
Computer follows them.**

INPUT rules + data

OUTPUT deterministic (fixed) result

MACHINE LEARNING

**You provide the examples.
Computer discovers the rules.**

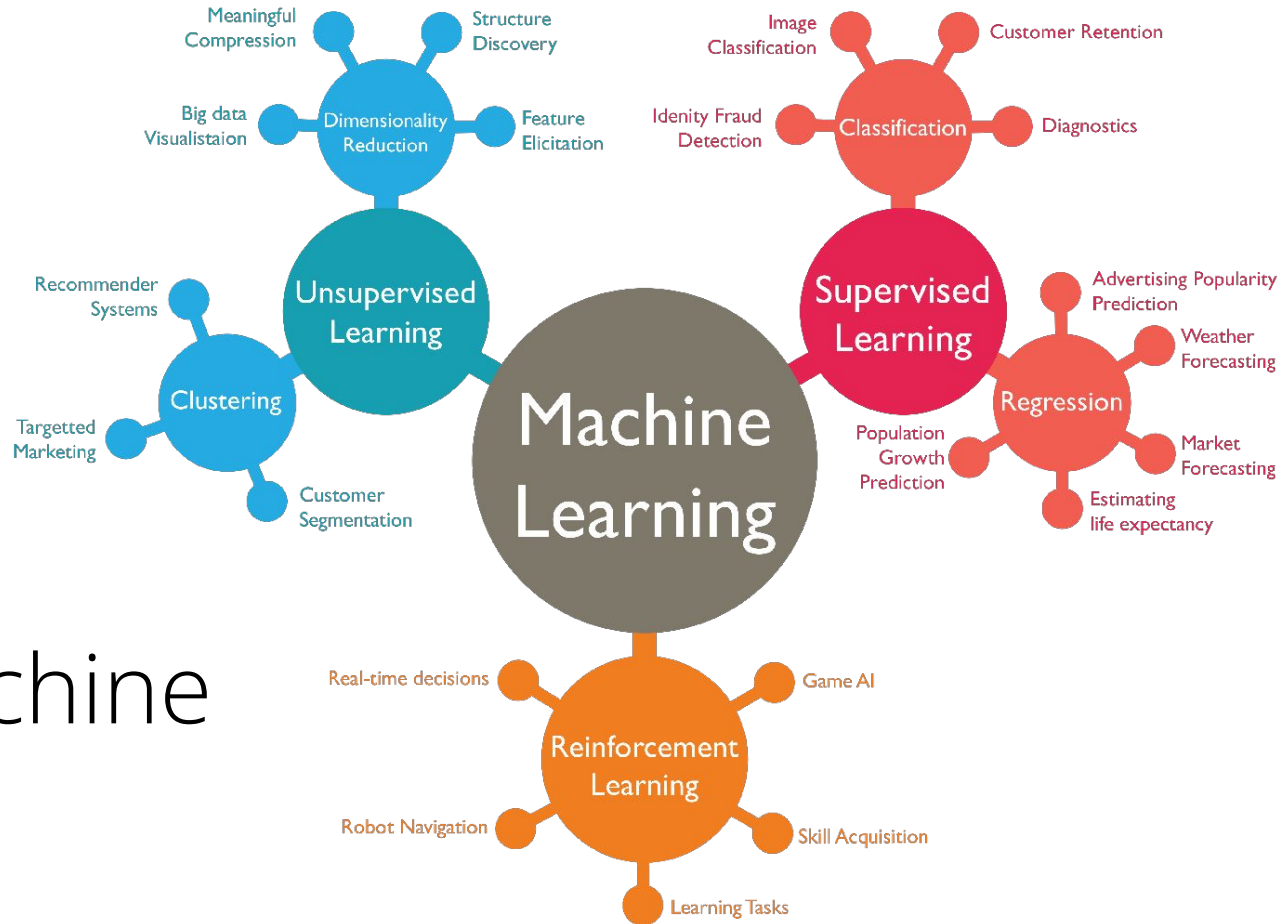
INPUT examples (data + correct answers)

OUTPUT prediction + confidence score

*"In traditional programming, you write the **rules**. In ML, you write the **examples** — and the computer figures out the rules for you. And often you don't even train it yourself — pre-trained models let you reuse examples someone else already learned from."*

Glossary

- Variable:** An attribute that represents a dimension of the data. Variables can be used as input features for models, and they can also be the output or target that the model is trying to predict. Variables can be categorical or numerical.
- Feature:** An individual measurable property or characteristic of a phenomenon being observed, often denoted as 'X' in models. Also known as *input*, *independent* or *predictor* variable.
- Label:** The variable we are predicting, often denoted as 'Y' in models. Also known as *output*, *dependent* or *target* variable.
- Algorithm:** A set of rules or procedures that a machine learning system follows to perform a task.
- Model:** The representation (internal model) learned by the machine learning algorithm.



Types of Machine Learning

X. THREE CATEGORIES OF ML

#1

Supervised Learning

PLAIN ENGLISH

You give the model input data and the correct answers. It learns patterns to predict answers for new, unseen data.

REAL-WORLD ANALOGY

A student studying past exam papers with answer keys. After practicing lots of questions, they learn the patterns and can solve brand-new questions on their own.

BUSINESS USE CASES

- Customer churn prediction
- Fraud detection
- Demand forecasting
- Sentiment and text classification
- Medical risk assessment
- Image labelling (e.g. telling cats apart from dogs)

Supervised Learning

Supervised Learning involves training a model on a set of *input-output pairs*. It learns a function that *maps* an input to an output based on example input-output pairs.

Features & Labels



Input (Feature)
e.g., email content



Output (Label)

Training Process



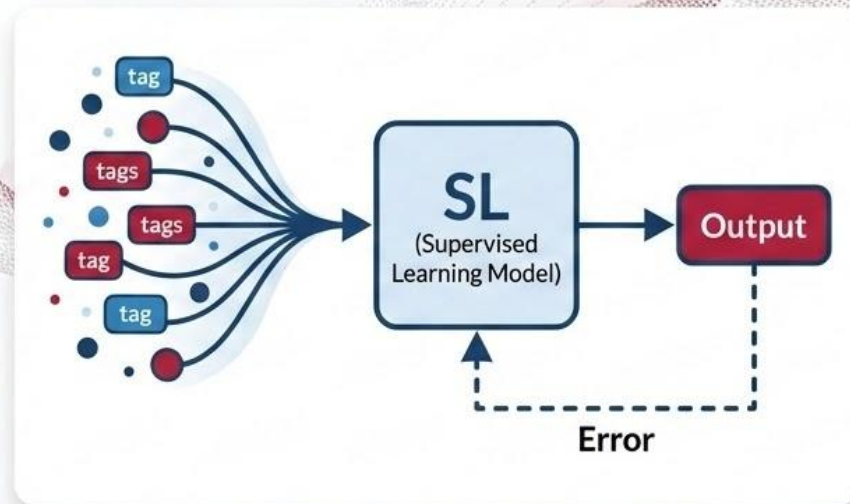
Labeled
Dataset



Algorithm Makes
Predictions



Compare with
Actual Labels (Error)



Supervised Learning (Types)

Aspect	Regression	Classification
Definition	Regression predicts a numerical output.	Classification predicts a categorical output.
Output Type	Numerical values (e.g., prices, temperatures).	Categorical labels (e.g., spam/not spam, dog/cat).
Nature of Predictions	Predicts a quantity that can take any value within a range.	Predicts a class label or category for the observation.
Algorithms	Linear Regression, Decision Trees, Random Forest, Support Vector Machines, Neural Networks, etc.	Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Neural Networks, etc.

Supervised Learning (Examples)



Regression	Classification
Predicting the price of a house based on its size, location, etc.	Determining if a tumor is benign or malignant based on medical imaging.
Forecasting tomorrow's temperature	Email spam detection

XI. THREE CATEGORIES OF ML

#2

Unsupervised Learning

PLAIN ENGLISH

You only give the model raw input data — no correct answers at all. The model finds hidden groups, patterns or unusual data points on its own.

REAL-WORLD ANALOGY

A librarian sorting a box of unmarked books. They group similar books together (romance, mystery, etc.) without anyone telling them the official genres.

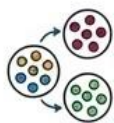
BUSINESS USE CASES

- Grouping customer into segments
- Anomaly detection in system logs
- Topic discovery in reviews or support tickets
- Building recommendation groups
- Preparing data for supervised learning models later on

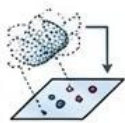
Unsupervised Learning

Unsupervised Learning involves training a model on a set of data without explicit instructions on what to look for. The goal is to identify patterns or structures in the data.

Clustering & Dimensionality Reduction



Clustering groups similar data points together.



Dimensionality Reduction reduces the number of random variables under consideration by obtaining a set of principal variables.

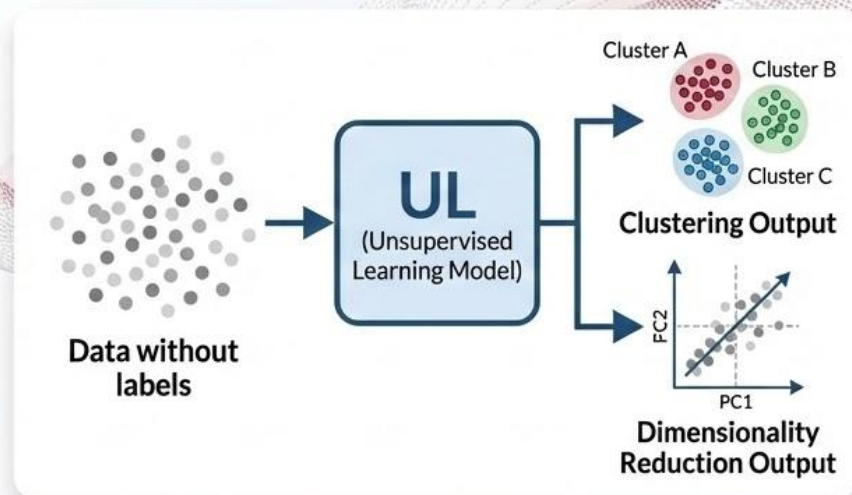
Training Process



The algorithm explores the data for patterns.



There's no 'correct answer' (labels) like in supervised learning.



XII. THREE CATEGORIES OF ML

#3

Reinforcement Learning

PLAIN ENGLISH

An AI agent acts within an environment. It gets rewards for good actions and penalties for bad ones. Over time, it learns the best way to act through trial and error.

REAL-WORLD ANALOGY

Training a dog with treats. Sit correctly → treat. Bark at the mailman → no treat. Over time the dog learns which actions pay off.

BUSINESS USE CASES

- AlphaGo, chess-playing AIs
- Robot control (warehouse pick-and-pack)
- Large recommendation systems
- RLHF training for modern LLMs

Reinforcement Learning

Reinforcement Learning is about taking suitable actions to *maximize reward* in a particular situation. It's trained using *trial and error*.

Agent, State, Reward



Agent

The decision maker.



State

The current situation returned by the environment.



Reward

Feedback from the environment.

Training Process

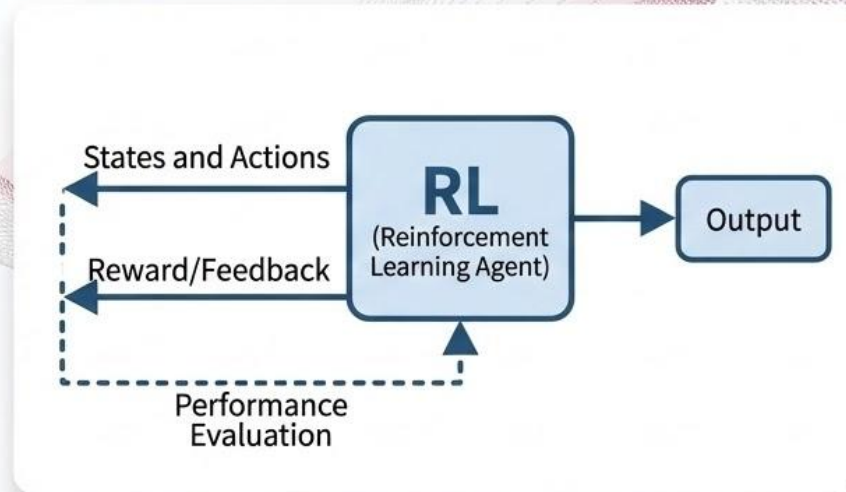


Trial and Error



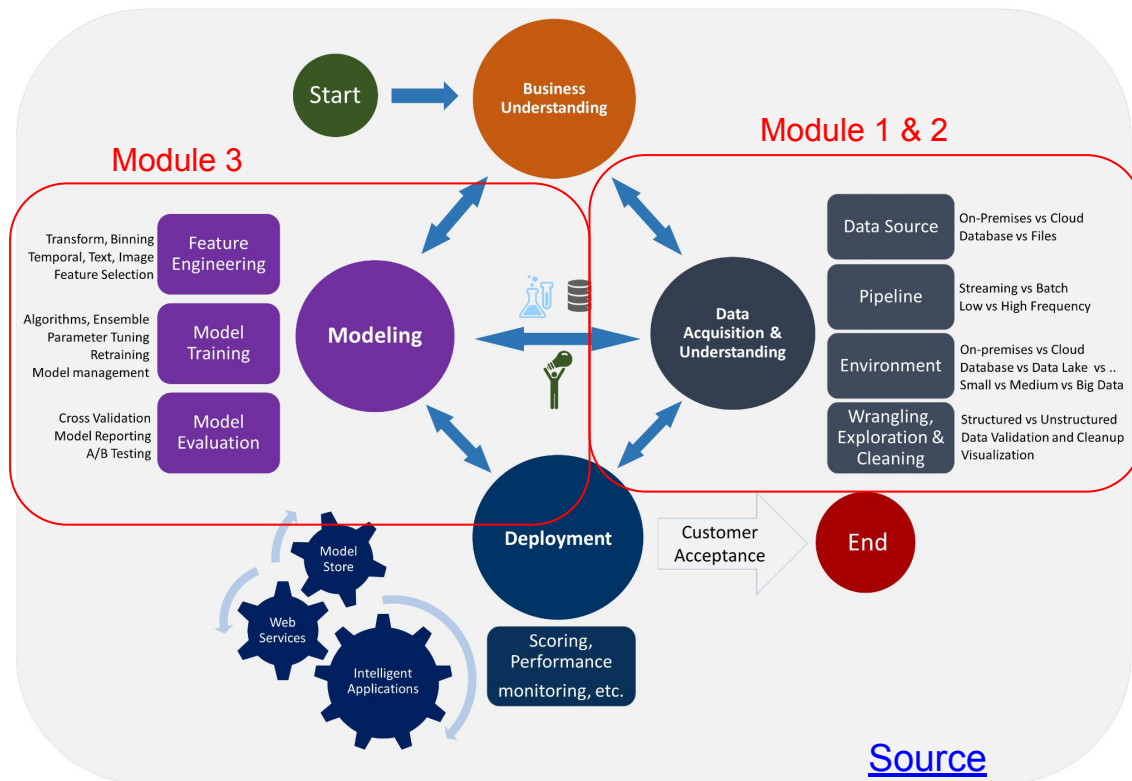
The agent takes actions; the environment returns the new state and a reward.

The goal is to find the best strategy, called **policy**, that will result in the cumulative reward for the agent.



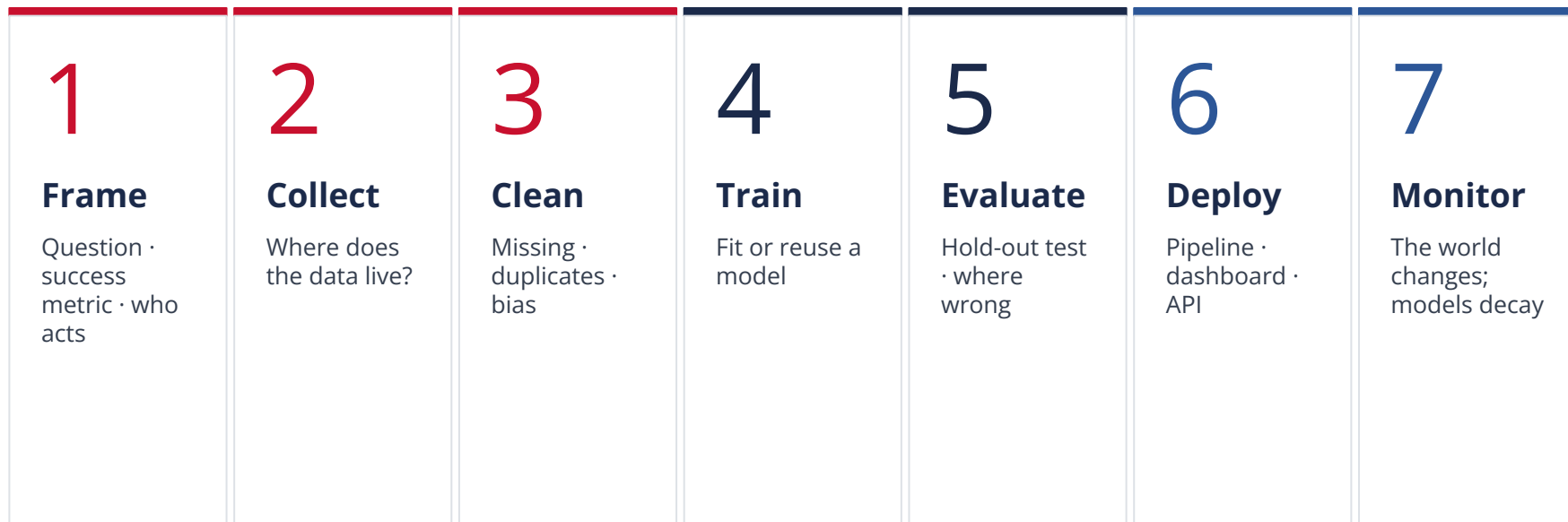
Data Science Lifecycle

1. Business understanding
2. Data acquisition and understanding
3. Modeling
4. Deployment
5. Customer acceptance



XIII. THE MAP EVERY REAL PROJECT FOLLOWS

Seven steps. In order. Every time.



■ Frame · Collect · Clean (≈65%)

■ Train · Evaluate (≈18%)

■ Deploy · Monitor (≈17%)

XVI. FRAMING IS THE ONLY STEP YOU CAN'T SKIP AND RECOVER FROM

*“A beautifully trained model that solves the wrong problem **is worth zero.**”*

The four questions every project must answer BEFORE touching data:

1

What is the question?

Be precise. ‘Reduce churn’ isn't a question. ‘Predict which customers cancel within 30 days’ is.

2

Who acts on the answer?

If nobody acts, no model matters. The action defines what ‘good’ means.

3

What's the success metric?

Define it before you build. Choose one. Resist adding more later.

4

What's the cost of being wrong?

Is a false positive worse than a false negative? Most domains have asymmetric costs.

Priya, after seeing Sarah's numbers...

“Sarah — the numbers look great.

But HOW SURE are we?

The model labeled every review — How do we know it isn't
confidently wrong *in ways we can't see from this report?”*

LESSON 02 · PROBABILITY & STATISTICS

Distribution · Confidence Intervals · Hypothesis Testing — all tools to help Sarah answer Priya's questions.

Come to L02 ready to dive in!

End of Lesson - Exit Ticket

Survey Link

Slido link will be provided in class

